

functions basic

```
In [1]: def greet():  
        print("Hello")  
        print("Good morning")  
        greet()
```

Hello
Good morning

```
In [2]: def greet():  
        print("Hello")  
        print("Good morning")  
        greet()  
  
def greet():  
    print("Hello")  
    print("Good morning")  
    greet()
```

Hello
Good morning
Hello
Good morning

```
In [3]: def greet():  
        print("Hello")  
        print("Good morning")  
        greet()  
        print()  
        def greet():  
            print("Hello")  
            print("Good morning")  
            greet()
```

Hello
Good morning

Hello
Good morning

```
In [4]: def greet():  
        print("Hello")  
        print("Good morning")  
        greet()  
        print('=====')  
        def greet():  
            print("Hello")  
            print("Good morning")  
            greet()  
        print('=====')  
        def greet():  
            print("Hello")  
            print("Good morning")  
            greet()
```

```

Hello
Good morning
=====
Hello
Good morning
=====
Hello
Good morning

```

```

In [5]: def greet():
        print("Hello")
        print("Good morning")
        greet()
        print()
        def greet():
            print("Hello")
            print("Good morning")
            greet()
            print()
        def greet():
            print("Hello")
            print("Good morning")
            greet()

```

```

Hello
Good morning

```

```

Hello
Good morning

```

```

Hello
Good morning

```

functions without arguments

```

In [6]: def greet():
        print('hello')
        print('good morning team')
        greet()

```

```

hello
good morning team

```

functions with argument

```

In [8]: def add(x,y):
        c = x+y
        print(c)
        add(5,5)

```

```

10

```

```

In [9]: def add(x,y):
        c = x+y
        return(c)#use return whenever you create a variable
        add(5,5)

```

Out[9]: 10

```
In [10]: def add(x,y):  
         c = x+y  
         return(c)#use return whenever you create a variable  
         add(5)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[10], line 4  
      2     c = x+y  
      3     return(c)#use return whenever you create a variable  
----> 4 add(5)  
  
TypeError: add() missing 1 required positional argument: 'y'
```

```
In [11]: def add(x,y):  
         c = x+y  
         return(c)#use return whenever you create a variable  
         add(5,5,6)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[11], line 4  
      2     c = x+y  
      3     return(c)#use return whenever you create a variable  
----> 4 add(5,5,6)  
  
TypeError: add() takes 2 positional arguments but 3 were given
```

```
In [12]: def add(x,y,z):  
         c=x+y  
         return c  
         add(5,5,2)
```

Out[12]: 10

```
In [13]: def add(x,y,z):  
         c=x+y+z  
         return c  
         add(5,5,2)
```

Out[13]: 12

```
In [14]: def add(x,y,z):  
         c=x+y+z+m  
         return c  
         add(5,5,2)
```

```

-----
NameError                                Traceback (most recent call last)
Cell In[14], line 4
      2     c=x+y+z+m
      3     return c
----> 4 add(5,5,2)

Cell In[14], line 2, in add(x, y, z)
      1 def add(x,y,z):
----> 2     c=x+y+z+m
      3     return c

NameError: name 'm' is not defined

```

```

In [15]: def add(x,y,z):
          c=x+y+z+m
          return c
          add(5,5,2,4)

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[15], line 4
      2     c=x+y+z+m
      3     return c
----> 4 add(5,5,2,4)

TypeError: add() takes 3 positional arguments but 4 were given

```

```

In [16]: def add(x,y,z,m):
          c=x+y+z+m
          return c
          add(5,5,2)

```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[16], line 4
      2     c=x+y+z+m
      3     return c
----> 4 add(5,5,2)

TypeError: add() missing 1 required positional argument: 'm'

```

```

In [17]: def add(x,y,z,m):
          c=x+y+z+m
          return c
          add(5,5,2,10)

```

Out[17]: 22

```

In [18]: def greet():
          print('Hello')
          print('good morning')
          greet()
          def add(x,y):
              c=x+y
              return c
          add(4,5)

```

Hello
good morning

Out[18]: 9

```
In [19]: def greet():
          print('Hello')
          print('good morning')
          greet()
          def add(x,y):
              c=x+y
              return c
          add(4,5)
          def sub(a,b):
              c = a-b
              return c
          sub(4,5)
```

Hello
good morning

Out[19]: -1

```
In [20]: def greet():
          print('Hello')
          print('good morning')

          def add(x,y):
              c=x+y
              return c

          def sub(a,b):
              c = a-b
              return c

          greet()
          add(5,4)
          sub(2,10)
```

Hello
good morning

Out[20]: -8

```
In [21]: def greet():
          print('Hello')
          print('good morning')

          def add(x,y):
              c=x+y
              return c

          def sub(a,b):
              c = a-b
              return c

          greet()
          print(add(5,4))
          print(sub(2,10))
```

Hello
good morning
9
-8

```
In [22]: def add_sub(x,y):
          c=x+y
          d=x-y
          return c,d
          add_sub(10,2)
```

Out[22]: (12, 8)

```
In [23]: def add_sub(x,y):
          c=x+y
          d=x-y
          return c,d
          add_sub(10,2)
          print(type(add_sub))
```

<class 'function'>

```
In [24]: def add_sub(x,y):
          c=x+y
          d=x-y
          return c,d
          result=add_sub(10,2)
          print(result)
          print(type(result))
```

(12, 8)

<class 'tuple'>

```
In [25]: def add_sub(x,y):
          c=x+y
          d=x-y
          return c,d
          result1,result2 = add_sub(10,2)
          print(result1)
          print(result2)
          print(type(result1))
```

12

8

<class 'int'>

```
In [30]: def add_sub_mul(x,y):
          c=x+y
          d=x-y
          e=x*y
          return c, d, e
          result1, result2, result3 = add_sub_mul(4,4)
          print('sum of 2 numbers:',result1)
          print('sub of 2 numbers:',result2)
          print('mul of 2 numbers:',result3)
          print(type(result1))
```

sum of 2 numbers: 8

sub of 2 numbers: 0

mul of 2 numbers: 16

<class 'int'>

UPDATE

```
In [31]: def update():  
         x = 8  
         print(x)  
         update()
```

8

```
In [32]: def update():  
         x = 8  
         print(x)  
         update(8)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[32], line 4  
      2     x=8  
      3     print(x)  
----> 4 update(8)  
  
TypeError: update() takes 0 positional arguments but 1 was given
```

```
In [34]: def update(x):#here the value of x=10 was just pointing but not updating  
         x = 8  
         return x  
         update(10)
```

Out[34]: 8

```
In [36]: def update(x):#to update we take another variable and call it  
         x = 8  
         return x  
         a = 10  
         update(a)  
         print(a)
```

10

```
In [37]: def update(x):#to update we take another variable and call it  
         x = 8  
         print(x)  
         a = 10  
         update(a)  
         print(a)
```

8

10

In []: